

Revised manuscript submission to Scientific Reports

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To: Dr Osman Orhan, Editorial Board Member

Dear Dr Orhan,

We present our revised manuscript, *Quantifying how spatial allocation and sampling alter population-weighted InSAR support summaries*, for reconsideration. The original title was *Auditing observability bias in InSAR-derived subsidence exposure estimates with a reproducible geospatial workflow*.

The revision focuses on a testable downstream accounting question: how much does spatial aggregation change a population-weighted InSAR support summary when the support field, domain and population total are fixed? Matched-input comparisons quantify the effect; within-cell covariance explains its mechanism; controlled missingness identifies both error directions and cases with small aggregate bias. Crossing GHSL and WorldPop with 28- and 224-pair selections preserves a positive difference across all 16 tested cases while demonstrating strong population-model dependence.

A connected LiCSBAS experiment separates sampled input support from the final quality mask and assesses internal reconstruction using excluded edges. Published Fresno observation polygons provide a second support definition. Added Census-block overlays, vertical-rate transfer and population-grid comparisons further test applicability and quantify important resolution and temporal limitations. Six further tests add actual irregular blocks and contiguous holes, spatial/date dependence, training-only quality curves, two regions, municipal residential eligibility and an archived independent-GNSS comparison. Small effects, incomplete coverage and negative physical-consistency results are retained as applicability limits.

We have withdrawn the interpretation of the original 3.77-million figure as a verified count of people exposed to subsidence in unobserved locations. The revised conclusion concerns allocation and aggregation effects; it does not establish deformation in missing locations. The Iran source attribution has also been corrected, and its incomplete analysis is excluded from quantitative applications.

The accompanying files contain the clean editable manuscript, Supplementary Information, separate main figures and a point-by-point response with page and line locations. An optional marked manuscript is explicitly limited to changes from a frozen local revision baseline. The scientific reproduction package and additional official source inputs are publicly available at the fixed GitHub release v0.5.0. The previous Zenodo DOI covers v0.3.0 only; a revised DOI deposit remains pending.

Sincerely,

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